



MODUL **PRAKTIKUM**

ADMINISTRASI BASIS DATA

Disusun Oleh:
Aprilia Kurnianti, S.T., M.Eng.
Dosen Pengampu Matkul ABD

Materi 7

Strategi Backup Data

A. Tujuan

Setelah mempelajari materi ini, mahasiswa diharapkan mampu:

- Mengubah Recovery Model
- Melakukan Full Backup
- Melakukan Differential Backup
- Memahami hubungan Full dan Differential
- Melakukan restore dengan prosedur yang benar
- Menganalisis hasil restore

B. Persiapan

- Gunakan database yang telah dibuat pada praktikum sebelumnya (DBKampus).
- Tabel utama: Mahasiswa (NIM, Nama, JenisKelamin, Prodi)

BAGIAN 1 – PERSIAPAN AWAL

- Langkah 1 – Pastikan Database Ada

BAGIAN 2 – RECOVERY MODEL

- Langkah 2 – Cek Recovery Model

```
/*Langkah 3 - Cek Recovery Model*/
SELECT name, recovery_model_desc
FROM sys.databases
WHERE name = 'DBKampus';
```

70 %

Results Messages

	name	recovery_model_desc
1	DBKampus	FULL

- Langkah 3 – Ubah ke FULL Recovery

```
/*Langkah 4 - Ubah ke FULL Recovery*/
ALTER DATABASE DBKampus
SET RECOVERY FULL;
```

70 %

Messages

Commands completed successfully.

Completion time: 2026-04-28T07:58:07.0548427+07:00

Cek kembali, Catatan penting :

Setelah mengubah ke FULL → WAJIB lakukan Full Backup pertama.

BAGIAN 3 – FULL BACKUP

- Langkah 4 – Pastikan Folder Backup Ada

Buat folder manual di:

D:\BackupSQL

Jika error “Cannot open backup device”, berarti:

- Folder belum ada
- SQL Server tidak punya permission

- Langkah 5 – Lakukan Full Backup

```
BACKUP DATABASE DBKampus
TO DISK = 'D:\BackupSQL\DBKampus_Full.bak'
WITH FORMAT,
    INIT,
    NAME = 'Full Backup DBKampus',
    STATS = 10;
```

70 %

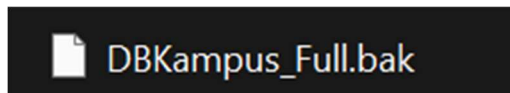
Messages

```
11 percent processed.
20 percent processed.
30 percent processed.
40 percent processed.
51 percent processed.
61 percent processed.
70 percent processed.
80 percent processed.
90 percent processed.
100 percent processed.
Processed 512 pages for database 'DBKampus', file 'DBKampus' on file 1.
Processed 1 pages for database 'DBKampus', file 'DBKampus_log' on file 1.
BACKUP DATABASE successfully processed 513 pages in 0.046 seconds (86.988 MB/sec).

Completion time: 2026-04-28T08:04:02.1778177+07:00
```

- Langkah 6 – Verifikasi File Backup

Cek di folder:



BAGIAN 4 – PERUBAHAN DATA (SIMULASI HARI KE-2)

- Langkah 7 – Tambahkan Data Baru

```

/*masuk ke database*/
USE DBKampus

INSERT INTO Akademik.Mahasiswa (NIM, Nama, TanggalLahir, Prodi, TahunMasuk)
VALUES
('20250140003', 'Budi', '2004-08-15', 'Teknologi Informasi', 2025),
('20250140004', 'Dewi', '2003-11-21', 'Teknologi Elektro', 2025);

```

70 %

Messages

(2 rows affected)

Completion time: 2026-04-28T08:12:28.8206853+07:00

- Langkah 8 – Verifikasi Data

```
SELECT * FROM Akademik.Mahasiswa;
```

70 %

Results Messages

	NIM	Nama	TanggalLahir	Prodi	TahunMasuk
1	20250140001	Andi	2004-02-10	Teknologi Informasi	2025
2	20250140002	Siti	2003-05-12	Teknologi Elektro	2025
3	20250140003	Budi	2004-08-15	Teknologi Informasi	2025
4	20250140004	Dewi	2003-11-21	Teknologi Elektro	2025

BAGIAN 5 – DIFFERENTIAL BACKUP

- Langkah 9 – Buat Differential Backup

```

/*eksekusi di master*/
BACKUP DATABASE DBKampus
TO DISK = 'D:\BackupSQL\DBKampus_Diff.bak'
WITH DIFFERENTIAL,
INIT,
NAME = 'Differential Backup DBKampus',
STATS = 10;

```

70 %

Messages

10 percent processed.
21 percent processed.
32 percent processed.
43 percent processed.
54 percent processed.
65 percent processed.
76 percent processed.
87 percent processed.
98 percent processed.
100 percent processed.
Processed 88 pages for database 'DBKampus', file 'DBKampus' on file 1.
Processed 1 pages for database 'DBKampus', file 'DBKampus_log' on file 1.
BACKUP DATABASE WITH DIFFERENTIAL successfully processed 89 pages in 0.014 seconds (49.211 MB/sec).
Completion time: 2026-04-28T10:01:19.4857545+07:00

Note: Differential hanya menyimpan perubahan sejak Full Back up terakhir.

BAGIAN 6 – SIMULASI KERUSAKAN DATA

- Langkah 10 – Simulasi Human Error

Misalnya terjadi kesalahan:

```

/*masuk ke database*/
DELETE FROM Akademik.Mahasiswa;

```

70 %

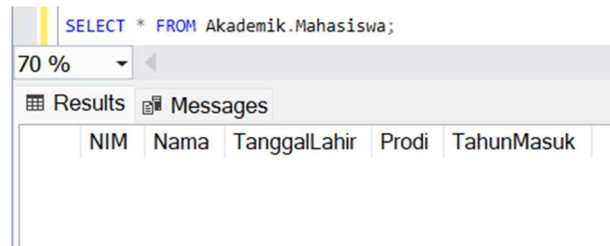
Messages

(4 rows affected)

Completion time: 2026-04-28T10:04:27.3044584+07:00

Semua data hilang!

Cek:



The screenshot shows a SQL query window with the text `SELECT * FROM Akademik.Mahasiswa;`. Below the query, there is a progress bar at 70%. The 'Results' tab is active, displaying a table with the following columns: NIM, Nama, TanggalLahir, Prodi, and TahunMasuk. The table is currently empty.

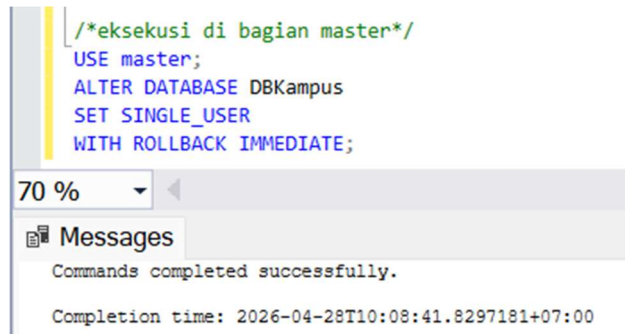
NIM	Nama	TanggalLahir	Prodi	TahunMasuk
-----	------	--------------	-------	------------

Hasil: 0 row.

BAGIAN 7 – PROSES RESTORE (PENTING!)

Pastikan database tidak sedang digunakan.

- Langkah 11 – Set Database ke Single User Mode



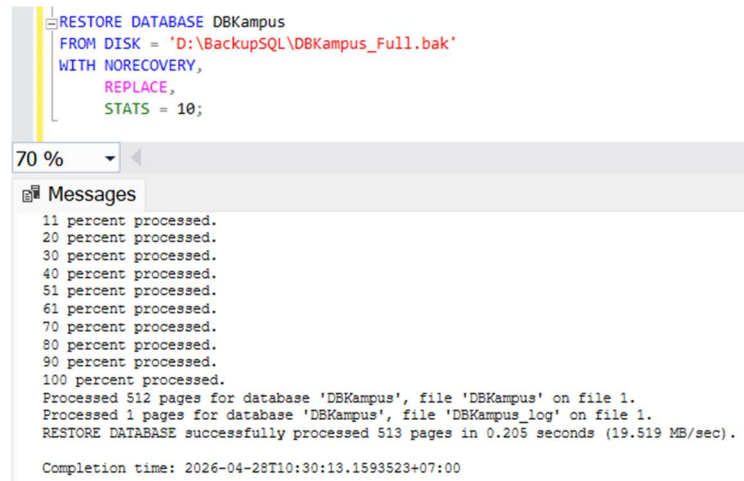
The screenshot shows a SQL query window with the following T-SQL script:

```
/*eksekusi di bagian master*/
USE master;
ALTER DATABASE DBKampus
SET SINGLE_USER
WITH ROLLBACK IMMEDIATE;
```

Below the script, there is a progress bar at 70%. The 'Messages' tab is active, displaying the following text:

Commands completed successfully.
Completion time: 2026-04-28T10:08:41.8297181+07:00

- Langkah 12 – Restore Full Backup



The screenshot shows a SQL query window with the following T-SQL script:

```
RESTORE DATABASE DBKampus
FROM DISK = 'D:\BackupSQL\DBKampus_Full.bak'
WITH NORECOVERY,
REPLACE,
STATS = 10;
```

Below the script, there is a progress bar at 70%. The 'Messages' tab is active, displaying the following text:

11 percent processed.
20 percent processed.
30 percent processed.
40 percent processed.
51 percent processed.
61 percent processed.
70 percent processed.
80 percent processed.
90 percent processed.
100 percent processed.
Processed 512 pages for database 'DBKampus', file 'DBKampus' on file 1.
Processed 1 pages for database 'DBKampus', file 'DBKampus_log' on file 1.
RESTORE DATABASE successfully processed 513 pages in 0.205 seconds (19.519 MB/sec).
Completion time: 2026-04-28T10:30:13.1593523+07:00

NORECOVERY artinya database belum aktif.

- Langkah 13 – Restore Differential Backup

```
RESTORE DATABASE DBKampus
FROM DISK = 'D:\BackupSQL\DBKampus_Diff.bak'
WITH RECOVERY,
STATS = 10;
```

70 %

Messages

10 percent processed.
21 percent processed.
32 percent processed.
43 percent processed.
54 percent processed.
65 percent processed.
76 percent processed.
87 percent processed.
98 percent processed.
100 percent processed.
Processed 88 pages for database 'DBKampus', file 'DBKampus' on file 1.
Processed 1 pages for database 'DBKampus', file 'DBKampus_log' on file 1.
DBCC execution completed. If DBCC printed error messages, contact your system administrator.
RESTORE DATABASE successfully processed 89 pages in 0.176 seconds (3.914 MB/sec).

Completion time: 2026-04-28T10:34:15.3677878+07:00

RECOVERY artinya database diaktifkan kembali.

- Langkah 14 – Kembalikan ke Multi User

```
ALTER DATABASE DBKampus
SET MULTI_USER;
```

70 %

Messages

Commands completed successfully.

Completion time: 2026-04-28T10:36:21.7648254+07:00

BAGIAN 8 – VERIFIKASI HASIL

- Langkah 15 – Cek Data

```
USE DBKampus;
SELECT * FROM Akademik.Mahasiswa;
```

70 %

Results **Messages**

	NIM	Nama	TanggalLahir	Prodi	TahunMasuk
1	20250140001	Andi	2004-02-10	Teknologi Informasi	2025
2	20250140002	Siti	2003-05-12	Teknologi Elektro	2025
3	20250140003	Budi	2004-08-15	Teknologi Informasi	2025
4	20250140004	Dewi	2003-11-21	Teknologi Elektro	2025

Hasil seharusnya kembali menjadi 4 data.

ANALISIS HASIL

1. Mengapa Full Backup harus direstore dulu?
2. Apa yang terjadi jika langsung restore Differential?
3. Mengapa menggunakan NORECOVERY pada Full?
4. Apa fungsi REPLACE?

TUGAS PRAKTIKUM LANJUTAN

1. Tambahkan data baru lagi
2. Buat Differential kedua
3. Hapus sebagian data (bukan semua)
4. Lakukan restore ulang
5. Analisis perbedaannya